

A Node-wise DOF restriction in the Extended Bearing Rigidity Matrix

Noyan Erdin Kilic

Abstract

The unified framework of [1] builds the extended bearing rigidity matrix as $\mathbf{B}_{\mathcal{G}}^+ = [\mathbf{D}_p \mathbf{U} \bar{\mathbf{E}}^\top \quad \mathbf{D}_a \mathbf{V} \bar{\mathbf{E}}_o^\top]$, with $\mathbf{U}$ and $\mathbf{V}$ indexed by edges, and characterises infinitesimal bearing rigidity by a rank condition. Working with heterogeneous formations we found that the edge-indexed factor cannot carry the motion constraints of both endpoints of an edge when those endpoints lie in different manifolds. We make this precise: requiring $\mathbf{U}_{ij}$ to be faithful to the bearing differential and to annihilate the motions the two agents cannot perform forces $\mathbf{D}_p^{(k)}(S_i - S_j) = 0$, which for a planar and a spatial agent holds only when the two are vertically aligned. Because the matrix is only determined on the admissible subspace, the choice made off that subspace does not affect the differential, but it does affect the rank, and the rank is what the rigidity condition compares. We give a four-agent formation on which the resulting test reports a rigid framework as flexible, propose a node-indexed form that removes the difficulty, and show that it reproduces the homogeneous construction of [1] exactly, so that no result for homogeneous formations is affected. All statements are checked against a finite-difference Jacobian of the bearing map.

1 Introduction

We came to this while using the framework of [1] to study topology design for formations whose agents live in different manifolds, for instance ground vehicles constrained to a plane together with aerial platforms in $\text{SE}(3)$. In that setting the rank condition of Theorem 2 in [1] sometimes reported frameworks as flexible that are in fact rigid, and the reason turned out to be structural rather than numerical.

Everything concerning homogeneous formations, in particular Table I and Section V of [1], is unaffected: the form proposed here reduces to it identically. The difficulty appears only for heterogeneous formations on incomplete graphs. Both case studies in [1] use complete graphs, where the rank condition holds trivially, so no conclusion drawn there changes either.

We believe that The general form of (10) in [1] cannot, in general, extend heterogeneous endpoint constraints off the admissible variation space in a way that simultaneously preserves the endpoint constraints and the null-space structure required by the rank argument of Theorem 2.

2 Setting

We follow the notation of [1]. A formation of $n \geq 3$ agents is a framework $(\mathcal{G}, \chi)$ with $\mathcal{G}$ a directed graph with m edges and $\chi \in \bar{\mathcal{D}} = \prod_i \mathcal{D}_i \subseteq \text{SE}(3)^n$. Agent i has position p_i , attitude R_i , and $c_i = \dim \mathcal{D}_i$ degrees of freedom; write $c = \sum_i c_i = \dim \bar{\mathcal{D}}$. An edge $e_k = (v_i, v_j)$ carries the bearing $b_{ij} = R_i^\top \hat{p}_{ij}$ with $\hat{p}_{ij} = (p_j - p_i)/\|p_j - p_i\|$ and $s_{ij} = \|p_j - p_i\|^{-1}$. Differentiating,

$$\dot{b}_{ij} = \underbrace{s_{ij} R_i^\top P(\hat{p}_{ij})}_{\mathbf{D}_p^{(k)}} (\dot{p}_j - \dot{p}_i) + \underbrace{(-R_i^\top [\hat{p}_{ij}]_\times)}_{\mathbf{D}_a^{(k)}} (-\omega_i), \quad (1)$$

with $P(x) = I_3 - xx^\top$ and ω_i the world-frame angular velocity. With the incidence matrices of [1] and $\bar{\mathbf{E}} = \mathbf{E} \otimes I_3$, $\bar{\mathbf{E}}_o = \mathbf{E}_o \otimes I_3$, one has $(\bar{\mathbf{E}}^\top x)_k = x_j - x_i$ and $(\bar{\mathbf{E}}_o^\top x)_k = -x_i$.

For each agent let S_i and P_i be the orthogonal projectors onto the translational and rotational directions it can realise:

$\mathcal{D}_i$	c_i	S_i	P_i
$\mathbb{R}^2$	2	$\text{diag}(1, 1, 0)$	0
$\mathbb{R}^3$	3	I_3	0
$\mathbb{R}^2 \times S^1$	3	$\text{diag}(1, 1, 0)$	$e_3 e_3^\top$
$\mathbb{R}^3 \times S^1$	4	I_3	$vv^\top$
$\text{SE}(3)$	6	I_3	I_3

Here $v \in S^2$ is the controllable axis of an $\mathbb{R}^3 \times S^1$ agent, and $\text{rk } S_i + \text{rk } P_i = c_i$. Following (8) of [1] a variation δ is lifted to $\delta^+ \in \mathbb{R}^{6n}$ by zero padding. Write

$$\mathcal{A} = \{\delta^+ : \delta \in \bar{\mathcal{I}}\} = \text{range } \bar{S} \oplus \text{range } \bar{P} \subseteq \mathbb{R}^{6n}, \quad (2)$$

with $\bar{S} = \text{blkdiag}(S_i)$, $\bar{P} = \text{blkdiag}(P_i)$, and let $A \in \mathbb{R}^{6n \times c}$ have orthonormal columns spanning $\mathcal{A}$. We call $\mathcal{A}$ the admissible subspace and $\mathcal{A}^\perp$ the virtual subspace, of dimension

$$q_v := 6n - c. \quad (3)$$

3 The two forms

Proposition 2 of [1] gives

$$\mathbf{B}_G^\times = \begin{bmatrix} \mathbf{D}_p \mathbf{U} \bar{\mathbf{E}}^\top & \mathbf{D}_a \mathbf{V} \bar{\mathbf{E}}_o^\top \end{bmatrix}, \quad \mathbf{U} = \text{diag}(\mathbf{U}_{ij}), \quad \mathbf{V} = \text{diag}(\mathbf{V}_{ij}). \quad (4)$$

We will compare it with

$$\mathbf{B}_G^\bullet = \begin{bmatrix} \mathbf{D}_p \bar{\mathbf{E}}^\top \bar{S} & \mathbf{D}_a \bar{\mathbf{E}}_o^\top \bar{P} \end{bmatrix}. \quad (5)$$

Acting on δ^+ , the k th row block of each is

$$(\mathbf{B}^\times \delta^+)_k = \mathbf{D}_p^{(k)} \mathbf{U}_{ij} (\delta_{p,j} - \delta_{p,i}) - \mathbf{D}_a^{(k)} \mathbf{V}_{ij} \delta_{a,i}, \quad (6)$$

$$(\mathbf{B}^\bullet \delta^+)_k = \mathbf{D}_p^{(k)} (S_j \delta_{p,j} - S_i \delta_{p,i}) - \mathbf{D}_a^{(k)} P_i \delta_{a,i}. \quad (7)$$

In (6) a single matrix multiplies the column blocks of both endpoints. In (7) each endpoint carries its own projector. For a homogeneous formation the two coincide, since $S_i = S_j$.

4 The matrix is determined only on $\mathcal{A}$

Lemma 1. *Relation (9) of [1] determines $\mathbf{B}_G^+$ on $\mathcal{A}$ and imposes nothing on $\mathcal{A}^\perp$.*

Proof. δ^+ ranges over $\mathcal{A}$, so (9) fixes $\mathbf{B}_G^+|_{\mathcal{A}}$. Entries acting on $\mathcal{A}^\perp$ are multiplied by zero. $\square$

Both (4) with Table III of [1] and (5) therefore satisfy Definition 13, and both reproduce (1) on admissible variations. They differ only on $\mathcal{A}^\perp$. That would be immaterial if only the differential were used, but Theorem 2 compares ranks, and the rank sees $\mathcal{A}^\perp$.

The text preceding Theorem 2 in [1] supplies the missing condition: the extended matrix differs from the ordinary one “because of the emergence of zero columns in correspondence of zero entries of the vector δ^+ ”, and q_v “corresponds to the number of null columns”. Stated without reference to a coordinate frame, the requirement is that the matrix vanish on the virtual subspace. We record the three properties the proof of Theorem 2 uses.

(R1) $\mathbf{B}_{\mathcal{G}}^+ A$ equals the differential (1) restricted to admissible variations.

(R2) $\mathbf{B}_{\mathcal{G}}^+(I - AA^\top) = 0$.

(R3) q_v is the same for $\mathcal{G}$ and for the complete graph $\mathcal{K}$, so that it cancels between $\text{rk } \mathbf{B}_{\mathcal{G}}^+ = 6n - q_v - q_i$ and $\text{rk } \mathbf{B}_{\mathcal{K}}^+ = 6n - q_v - q_t$.

Remark 2. (R2) is equivalent to “every inadmissible coordinate indexes a null column” only when $\mathcal{A}$ is spanned by coordinate directions. This holds for $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^2 \times S^1$ and $\text{SE}(3)$, and for $\mathbb{R}^3 \times S^1$ under the parametrisation of Table I, which stores $\dot{\theta}$ in the third slot of $\delta_{a,i}$ and maps it through $\mathbf{V}_{ij} = [0_{3 \times 2} \ v]$. It does not hold if one instead writes $P_i = vv^\top$ and keeps ω_i itself as the variable, since $vv^\top$ has rank one but no zero column unless v is a coordinate axis. The two parametrisations give the same rank and the same q_v ; only the column-counting phrasing is sensitive to the choice, which is why we state (R2) as an identity.

Remark 3. Definition (3) takes q_v to be structural rather than a count of null columns of $\mathbf{B}_{\mathcal{G}}^+$, and (R3) then holds by construction. The two disagree on sparse graphs: if agent i is isolated in $\mathcal{G}$, all six of its columns vanish, including the c_i admissible ones, and those extra zeros are infinitesimal variations of $\mathcal{G}$ belonging to q_i .

5 An edge-indexed factor cannot satisfy (R1) and (R2)

Theorem 4. Let $e_k = (v_i, v_j)$ and suppose (4) satisfies (R1) and (R2) on the column blocks of i and j . Then

$$\mathbf{D}_p^{(k)}(S_i - S_j) = 0, \quad \text{that is} \quad \text{range}(S_i - S_j) \subseteq \text{span}\{\hat{p}_{ij}\}. \quad (8)$$

If $S_i \neq S_j$ this fails outside a set of configurations of measure zero.

Proof. Write $U = \mathbf{U}_{ij}$. By (6) the column block of $\mathbf{B}^\times$ in row block k is $-\mathbf{D}_p^{(k)}U$ for agent i and $+\mathbf{D}_p^{(k)}U$ for agent j . Comparing with (1) on admissible directions, (R1) gives

$$\mathbf{D}_p^{(k)}US_i = \mathbf{D}_p^{(k)}S_i, \quad \mathbf{D}_p^{(k)}US_j = \mathbf{D}_p^{(k)}S_j,$$

and (R2) applied to the inadmissible directions of each endpoint gives

$$\mathbf{D}_p^{(k)}U(I_3 - S_i) = 0, \quad \mathbf{D}_p^{(k)}U(I_3 - S_j) = 0.$$

Adding the two conditions for i yields $\mathbf{D}_p^{(k)}U = \mathbf{D}_p^{(k)}S_i$, and those for j yield $\mathbf{D}_p^{(k)}U = \mathbf{D}_p^{(k)}S_j$; subtracting gives (8). Since $\mathbf{D}_p^{(k)} = s_{ij}R_i^\top P(\hat{p}_{ij})$ with $s_{ij} \neq 0$ and R_i orthogonal, $\ker \mathbf{D}_p^{(k)} = \text{span}\{\hat{p}_{ij}\}$, which gives the second form. If $S_i \neq S_j$ then $S_i - S_j$ is a nonzero symmetric matrix and (8) confines its range to a single line, a nontrivial algebraic condition on (p_i, p_j) . $\square$

Corollary 5. For i planar and j spatial, $S_i - S_j = -e_3e_3^\top$ and (8) reads $P(\hat{p}_{ij})e_3 = 0$, so the two agents must be vertically aligned. Table III of [1] takes $\mathbf{U}_{ij} = I_3$ on such edges, which satisfies (R1) but not (R2): the column of agent i for its z coordinate is $-\mathbf{D}_p^{(k)}e_3 \neq 0$, although a planar agent cannot translate in z .

6 Consequences

Proposition 6. Under (4) with $\mathbf{U}_{ij} = I_3$ on mixed edges, the z column of a planar agent vanishes exactly when that agent is incident to no spatial agent. The number of null columns therefore depends on $\mathcal{E}$, and (R3) fails.

Since q_v no longer cancels between $\mathcal{G}$ and $\mathcal{K}$, the equivalence in the proof of Theorem 2 of [1] does not go through. The following formation shows that the conclusion can fail as well.

Example 7. Let $n = 4$ with $\mathcal{D}_1 = \mathcal{D}_2 = \mathcal{D}_3 = \mathbb{R}^2$ and $\mathcal{D}_4 = \mathbb{R}^3$, so $c = 9$ and $q_v = 15$, at

$$p_1 = (0, 0, 0), \quad p_2 = (1, 0, 0), \quad p_3 = (1, 1, 0), \quad p_4 = (\tfrac{1}{2}, \tfrac{1}{2}, 1), \quad R_i = I_3,$$

and let $\mathcal{G}$ have the four edges $\{(1, 2), (1, 3), (2, 4), (3, 4)\}$.

	rk $\mathbf{B}_{\mathcal{G}}$	rk $\mathbf{B}_{\mathcal{K}}$	conclusion
edge-indexed (4)	6	8	flexible
node-indexed (5)	6	6	rigid
finite differences on $\bar{\mathcal{I}}$	6	6	rigid

The last row differentiates the bearing map along admissible variations and uses neither construction. The framework admits two translations and a uniform scaling and nothing else, so $c - q_t = 9 - 3 = 6$ is full rank on $\bar{\mathcal{I}}$; $\mathcal{G}$ attains it with four edges and is minimally rigid.

Remark 8. For $\mathcal{K}$ in Example 7 the edge-indexed matrix has $\text{rk} = 8$, and deleting the three z columns of the planar agents returns it to 6. Those columns carry two dimensions of rank that correspond to no admissible motion. One combination of them is a null direction, namely the global z translation, which the planar agents cannot perform. The same effect appears in the case study of Section VI-B of [1]: there $q_t = 5$ is reported while the accompanying list of trivial motions has four entries, the difference being the global z translation of a formation whose unicycles are confined to a plane. The parenthetical remark that columns 3, 6, 9, 10, 11 are linearly dependent on the remainder is true of each column separately, but the three z columns are not jointly removable: deleting them lowers the rank from 13 to 11.

7 The node-indexed form

Theorem 9. $\mathbf{B}_{\mathcal{G}}^{\bullet}$ satisfies (R1), (R2) and (R3), and for every $\mathcal{G}$

$$\text{rk } \mathbf{B}_{\mathcal{G}}^{\bullet} = 6n - q_v - q_i, \quad q_i = \dim \ker(\mathbf{B}_{\mathcal{G}}^{\bullet} A). \quad (9)$$

Consequently $\text{rk } \mathbf{B}_{\mathcal{G}}^{\bullet} = \text{rk } \mathbf{B}_{\mathcal{K}}^{\bullet}$ holds if and only if $q_i = q_t$, that is, if and only if $(\mathcal{G}, \chi)$ is IBR.

Proof. (R1) follows by substituting (7) and using $S_i \delta_{p,i} = \delta_{p,i}$, $P_i \delta_{a,i} = \delta_{a,i}$ for $\delta^+ \in \mathcal{A}$. For (R2), $\bar{S}$ and $\bar{P}$ are the orthogonal projectors onto the admissible subspace, so $\mathbf{B}^{\bullet}(I - AA^{\top}) = 0$; this depends only on the agents' manifolds and not on $\mathcal{E}$, which is (R3) with (3).

For (9), (R2) gives $\mathbf{B}_{\mathcal{G}}^{\bullet} = \mathbf{B}_{\mathcal{G}}^{\bullet} AA^{\top}$, hence $\text{range } \mathbf{B}_{\mathcal{G}}^{\bullet} = \text{range}(\mathbf{B}_{\mathcal{G}}^{\bullet} A)$ and

$$\text{rk } \mathbf{B}_{\mathcal{G}}^{\bullet} = \text{rk}(\mathbf{B}_{\mathcal{G}}^{\bullet} A) = c - \dim \ker(\mathbf{B}_{\mathcal{G}}^{\bullet} A) = (6n - q_v) - q_i.$$

The same applies to $\mathcal{K}$ with $q_t = \dim \ker(\mathbf{B}_{\mathcal{K}}^{\bullet} A)$, and q_v is the same constant in both. By Theorem 1 of [1], $\ker \mathbf{B}_{\mathcal{K}} \subseteq \ker \mathbf{B}_{\mathcal{G}}$, so equality of dimensions is equality of the two kernels, which is Definition 12 of [1]. $\square$

Proposition 10. If $\mathcal{D}_i = \mathcal{D}$ for all i and $\mathbf{U}_{ij} = S$, $\mathbf{V}_{ij} = P$ are the Table I entries, then $\mathbf{B}_{\mathcal{G}}^{\times} = \mathbf{B}_{\mathcal{G}}^{\bullet}$ for every graph and configuration.

Proof. $(\mathbf{U}\bar{\mathbf{E}}^{\top}x)_k = S(x_j - x_i) = Sx_j - Sx_i = (\bar{\mathbf{E}}^{\top}\bar{S}x)_k$ since $\bar{S} = I_n \otimes S$, and likewise $(\mathbf{V}\bar{\mathbf{E}}_o^{\top}y)_k = -Py_i = (\bar{\mathbf{E}}_o^{\top}\bar{P}y)_k$. $\square$

For $\mathbb{R}^3 \times S^1$, Proposition 10 is read with $\mathbf{V}_{ij}$ in the projector parametrisation of Remark 2; the two parametrisations differ as matrices but not in rank or in q_v .

In implementation terms the change is to move $\mathbf{U}$ and $\mathbf{V}$ from between $\mathbf{D}_p, \mathbf{D}_a$ and the incidence matrices to the far right, as node-indexed block-diagonal projectors. Per row block $k = (i, j)$, with each index naming the top-left corner of a 3×3 block,

$$\begin{aligned}\mathbf{B}^\bullet[3k, 3j] &+= \mathbf{D}_p^{(k)} S_j, \\ \mathbf{B}^\bullet[3k, 3i] &-= \mathbf{D}_p^{(k)} S_i, \\ \mathbf{B}^\bullet[3k, 3n+3i] &+= \mathbf{D}_a^{(k)} P_i.\end{aligned}\tag{10}$$

The two dense $3m \times 3m$ factors $\mathbf{U}$ and $\mathbf{V}$ are no longer needed.

8 Numerical checks

The reference in what follows is the Jacobian of the bearing map with respect to admissible variations, obtained by central differences on the manifold, with attitudes updated as $R_i \mapsto \exp([\varepsilon\omega_i]_\times)R_i$. It uses neither construction.

Over 432 frameworks covering the five manifolds at $n = 3, 4, 6, 9$ and sixteen heterogeneous mixes at four graph densities each, with randomised $\mathbb{R}^3 \times S^1$ axes, (5) reproduces this Jacobian on $\mathcal{A}$ to a worst relative error of 1.1×10^{-9} . So does (4), as Lemma 1 requires, once each form is read in its own parametrisation of the $\mathbb{R}^3 \times S^1$ rotational coordinate (Remark 2). The identity $\mathbf{B}^\bullet(I - AA^\top) = 0$ holds to 2.2×10^{-16} .

Over 1200 random heterogeneous frameworks with $n \in \{4, 5, 6\}$:

	edge-indexed	node-indexed
(9) violated	517/1200	0/1200
rank condition disagrees with IBR	81/1200 (6.8%)	0/1200

For homogeneous formations the two constructions agree bitwise, checked over 25 graphs in each manifold at $n = 3, 4, 6, 8, 12$, as Proposition 10 predicts. The homogeneous ranks match the closed forms $2n - 3, 3n - 4, 3n - 4, 4n - 5, 6n - 7$ at $n = 3, 4, 6, 8, 12, 16$, and for a heterogeneous mix $\text{rk } \mathbf{B}_K \leq c - q_t$ holds in every case tested. Reproducing the case study of Section VI-B of [1] with (4) returns $\text{rk } \mathbf{B}_K^+ = 13$ with six null columns, the values reported there, which indicates that the difference lies in the construction and not in our implementation of it. Deleting the three z columns of the unicycles there lowers the rank from 13 to 11, and of the five motions listed as trivial only four are admissible.

9 Discussion

Where the ambiguity comes from

For a homogeneous planar formation, $\text{diag}(1, 1, 0)$ serves two purposes at once. It restricts the admissible translations, and it records that a bearing between two coplanar agents has no z component, so that the measurement manifold is $S^1 \subset S^2$. Table I is consistent with both readings and nothing separates them.

They separate as soon as $S_i \neq S_j$. On an edge between a planar and a spatial agent the bearing does have a nonzero z component, so the measurement reading gives $\mathbf{U}_{ij} = I_3$, which is the Table III entry, while the variation reading asks for two different restrictions at the two endpoints. The proof of Proposition 2 in [1] describes $\mathbf{U}$ as embedding the variation of each bearing in S^2 , which is the first reading, whereas the rank accounting behind Theorem 2 needs the second. Separating the two is what (5) does (Theorem 9): $\mathbf{D}_p^{(k)}$ already carries the measurement geometry, and $\tilde{S}, \tilde{P}$ carry the motion constraints.

Possible next steps

The observation is small on its own. It becomes more interesting in what it enables, and we would be glad to hear whether any of the following seems worth pursuing jointly.

Minimality is the first candidate. The counting arguments that give the minimum number of edges for rigidity in $\mathbb{R}^d$ rely on every edge contributing the same rank. Once endpoints carry different constraints the per-edge contribution varies with the pair of manifolds, and the corresponding bounds for heterogeneous formations do not appear to be settled.

The second is the design problem that led us here: choosing which bearings to maintain in a heterogeneous formation so that the framework is rigid with as few measurements as possible. Sparsity alone underdetermines the answer, since many edge sets of the same minimal size are rigid at the same configuration and differ considerably in the rigidity eigenvalue.

References

- [1] G. Michieletto, A. Cenedese, and D. Zelazo, “A Unified Dissertation on Bearing Rigidity Theory,” *IEEE Transactions on Control of Network Systems*, vol. 8, no. 4, pp. 1624–1636, Dec. 2021. [doi:10.1109/TCNS.2021.3077712](https://doi.org/10.1109/TCNS.2021.3077712)